



STM42M SERIES

Integrated Step-servo Motor Instruction Manual



Beijing NiMotion Control Technology Co., Ltd

Safety Instructions

 Danger (May cause death or serious injury!)	
Wiring job, be sure to by professional electrical engineer according to the connection diagram installation operation.	Danger of electric shock and fire.
Before wiring, make sure that the power supply is off.	Danger of electric shock and fire.
The ground wire of the motor and the input terminal of the power must be firmly installed.	Danger of electric shock and fire.
Don't bend, pull or clamp the connecting cable forcibly.	Danger of fire.
When using on the lifting device, please take measures to protect the position of the movable part.	Danger of injury.
Please set up circuit breakers and other safety devices to prevent external circuit short circuit and equipment failure can cut off the power supply in time.	Danger of electric shock and fire.
Don't use it in places with severe vibration and impact.	Danger of fire.
After the driver is powered off, the bus voltage will remain for a period of time. Don't touch the power input terminal within two minutes.	Danger of electric shock and fire.
For the power supply of the motor, please use the primary and secondary reinforced insulated DC power supply.	Danger of electric shock.
Please cut off the power supply of the motor during power failure, otherwise the motor will start suddenly after restoring the power supply.	Danger of injury.
When the motor is running, please ensure safety before releasing the holding force of the motor.	Danger of injury.
Don't use the motor near inflammable, explosive, corrosive environment and places easy to touch water.	Danger of electric shock and fire.
 Warning (May result in injury or damage to objects!)	
When using the motor, don't exceed its specification value.	Danger of injury and motor fault.
Don't hold the motor output shaft or cable during handling.	Danger of injury and motor fault.
Please install the protective cover on the rotating part (output shaft) of the motor.	Danger of injury.
Don't stack obstacles hindering ventilation around the motor to ensure heat dissipation.	Danger of motor fault.
Don't touch the motor when it is running or within a short time after it stops running. Please post a warning sign in a conspicuous position.	Danger of scalding.
Please use the motor according to the specified parameters.	Danger of motor fault and fire.
Please install an emergency stop device or emergency stop circuit outside the motor to ensure safe shutdown in case of failure or abnormal operation.	Danger of injury.
Please don't touch the shell during insulation resistance measurement or insulation withstand voltage test.	Danger of electric shock.
When the motor is abnormal, please stop running immediately and cut off the power supply to avoid fire or injury.	Danger of injury and fire.
When using, ensure that the grounding resistance of the system is less than 4Ω. When multiple motors are used in parallel, the grounding resistance between the ground of each	Danger of motor fault.

motor is less than or equal to 0.1Ω. The RS485 interface of the master RS485 controller and the slave station of the integrated step-servo motor should be in common ground, and the grounding resistance should be less than or equal to 0.1Ω. The motor shell shall be installed on the motor bracket through the installation hole. The bracket shall be reliably connected to the ground, and the grounding resistance shall be less than or equal to 0.1Ω.	
 Caution	
Don't disassemble or modify the motor, otherwise it may cause injury. For inspection or repair, please contact Beijing NiMotion Control Technology Co., Ltd.	
When the motor is scrapped, please disassemble it as far as possible and treat it as industrial waste.	
Before use, please read and fully understand the "Safety Instructions" to use the motor correctly.	
This product is designed and manufactured for use in general industrial equipment. Don't use it for any other purpose. We hereby state that we will not be liable for any damages caused by ignoring this advice.	

1. Summary

1.1 Preface

Thank you for using STM42M series integrated step-servo motor products of Beijing NiMotion Control Technology Co., Ltd! Please read this manual carefully before using!

STM42M series integrated step-servo motor products are high-performance and low-power step-servo products independently developed by our company. With its excellent closed-loop control system and integrated development concept, integrated stepper motor, encoder, driver in one. The product adopts FOC magnetic field oriented control technology and SVPWM technology, and supports RS485 or CANopen bus control, has the characteristics of small size, simple wiring etc. Step-servo motor has both the functions of servo motor and stepping motor, and the price is only the level of stepping motor, which can provide a cost-effective solution for customers who need the performance of servo motor. We hope that our products can help you achieve the desired motion control with superior performance, excellent quality and outstanding cost performance.

If there is any change to this manual, please refer to the new version without further notice.

1.2 Version Description

Version	Description of Revised Content	Date
A	Establish	2023-3-16

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2. Product Inspection and Model Description

2.1 Unpacking Inspection

Please check the following items when unpacking:

- Whether it is the ordered product: Please check that the model and specification on the nameplate of the motor is the same as your order.
- Whether the product is damaged: Visually inspect the appearance for damage or scratches.
- Whether the screws are loose: Whether any screws are not locked or fall off.
- Motor without brake check whether the shaft is running smoothly: Rotate the motor shaft by hand, if it can run smoothly, it means that the motor shaft is normal.
- Motor with brake check whether the brake is normal: If the motor shaft is locked, the brake is normal.
- In case of any of the above situations, please contact us for proper solution.

Complete operational product components shall be included as follows:

- Integrated step-servo motor.
- Communication cable required.
- Communication interface components required.

2.2 Product Model Comparison

2.2.1 Naming Rules for Product Model

STM 42 XX B - 485 - M - 0 F S B												
①	②	③	④	⑤	⑥	⑦	⑧	⑨	⑩			
① STM: Integrated Stepper Motor					② Width of motor base: 42-42mm							
③ Motor length: 34-34mm;40-40mm;48-48mm;60-60mm												
④ Performance of encoder: B-2500 lines												
⑤ Communication mode: CANopen-CANopen; 485-RS485						⑥ Hardware feature code						
⑦ Length of shaft: 0-Standard length; 1-Non standard length												
⑧ Shaft key form: S-Light shaft; F-Milling flat; K-Half circle key; H-Flat key; M-Perforated installation												
⑨ Special processing code: S-No special processing; A-Gear; B-Screw; E-Center thread hole...												
⑩ Brake: Default-Without break; B-With break												

Figure1 Naming Rules for STM42M Series

2.2.2 Installation Dimension of Motor

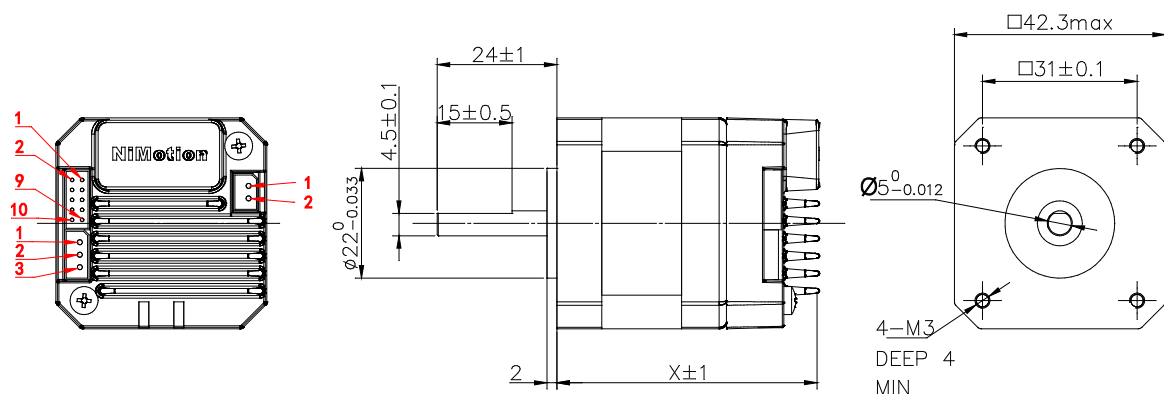


Figure2 Structure Diagram of Motor without brake

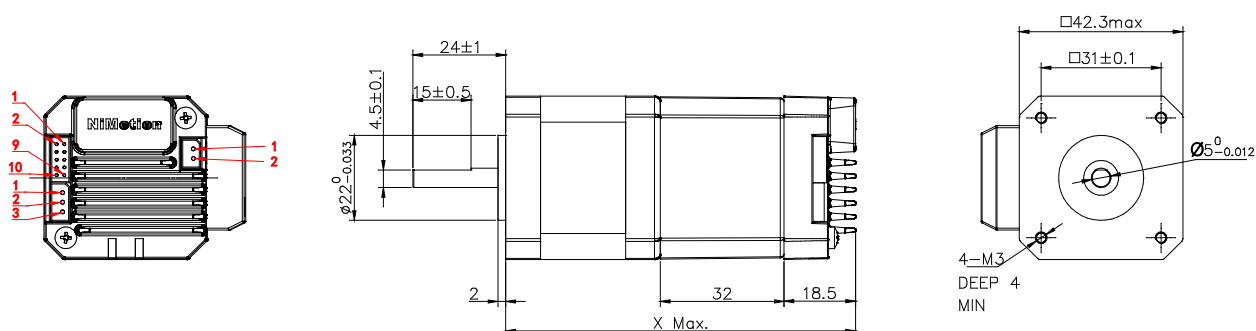


Figure3 Structure Diagram of Motor without brake

Table1 Motor Model and Some Key Dimensions

Model	Length (X in Figure) (mm)	Diameter of shaft	With or without brake	Holding torque of brake(N·m)	Rotor inertia (kg·m ²)	Weight (kg)
STM4234B- CANopen/485-M-0FS	52.5 (51.5)	Φ5	No	-	3.4*10 ⁻⁶	0.27
STM4240B- CANopen/485-M-0FS	58.5 (57.5)	Φ5	No	-	5.4*10 ⁻⁶	0.33
STM4248B- CANopen/485-M-0FS	66.5 (65.5)	Φ5	No	-	6.8*10 ⁻⁶	0.41
STM4260B- CANopen/485-M-0FS	78.5 (77.5)	Φ5	No	-	8*10 ⁻⁶	0.52
STM4234B- CANopen/485-M- 0FSB	84.5 (82.5)	Φ5	Yes	0.65	3.4*10 ⁻⁶	0.46

STM4240B- CANopen/485-M- 0FSB	90.5 (88.5)	Φ5	Yes	0.65	5.4×10^{-6}	0.52
STM4248B- CANopen/485-M- 0FSB	98.5 (96.5)	Φ5	Yes	0.65	6.8×10^{-6}	0.60
STM4260B- CANopen/485-M- 0FSB	109.5 (107.5)	Φ5	Yes	0.65	8×10^{-6}	0.71

Note: The dimensions in parentheses are the dimensions of the anti-collision structure motor.

2.3 Motor Torque Curve

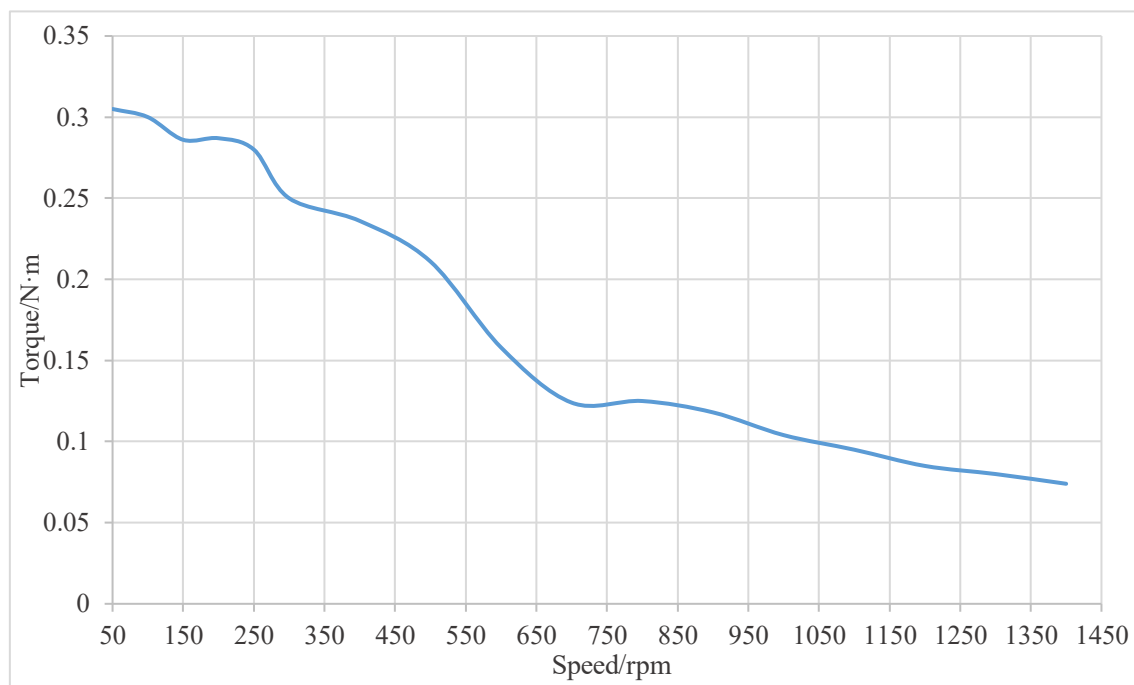


Figure4 STM4234B-CANopen/485-M-XX Frequency-Torque Characteristics(@24V, 1.3A)

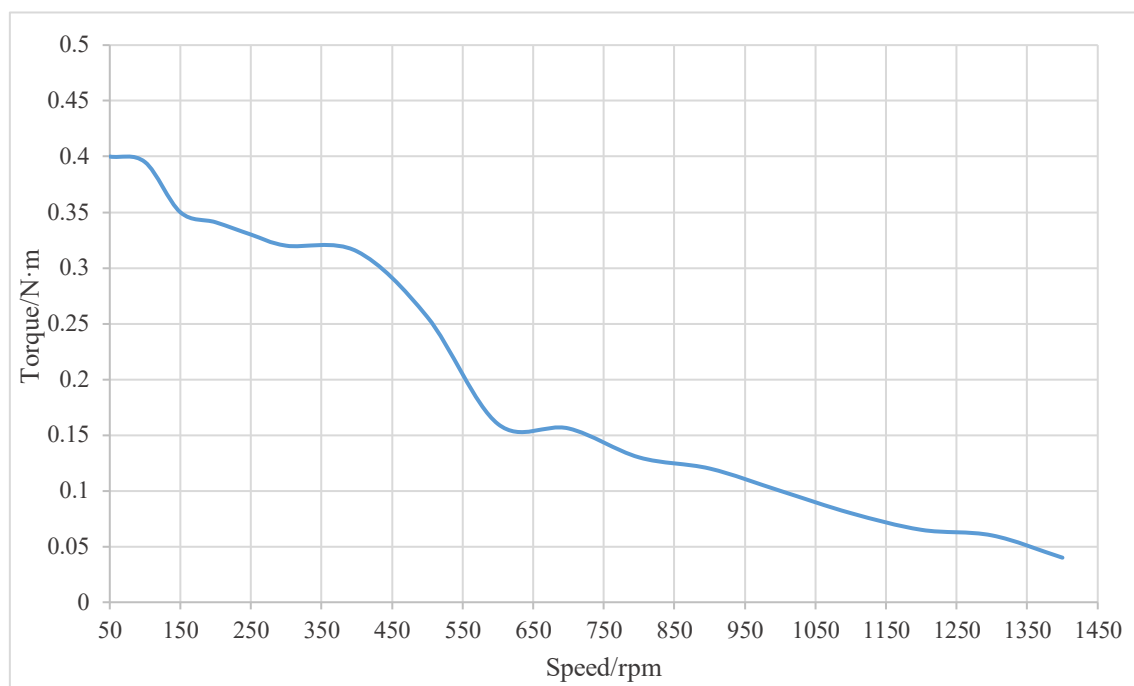


Figure5 STM4240B-CANopen/485-M-XX Frequency-Torque Characteristics(@24V, 1.7A)

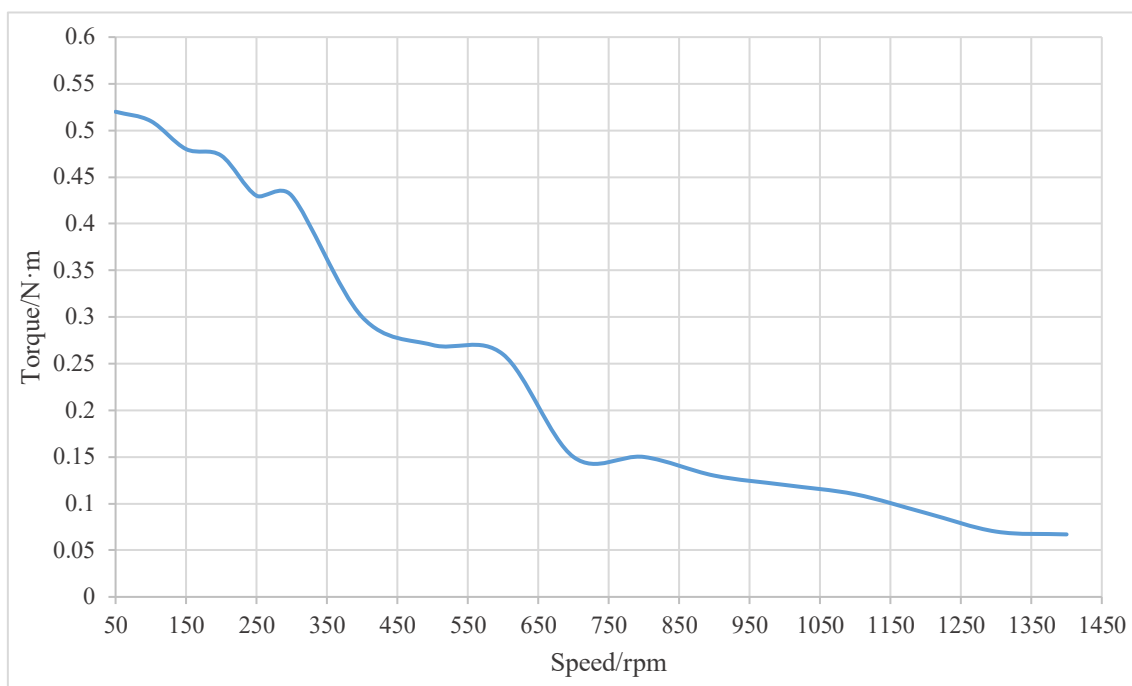


Figure6 STM4248B-CANopen/485-M-XX Frequency-Torque Characteristics(@24V, 1.7A)

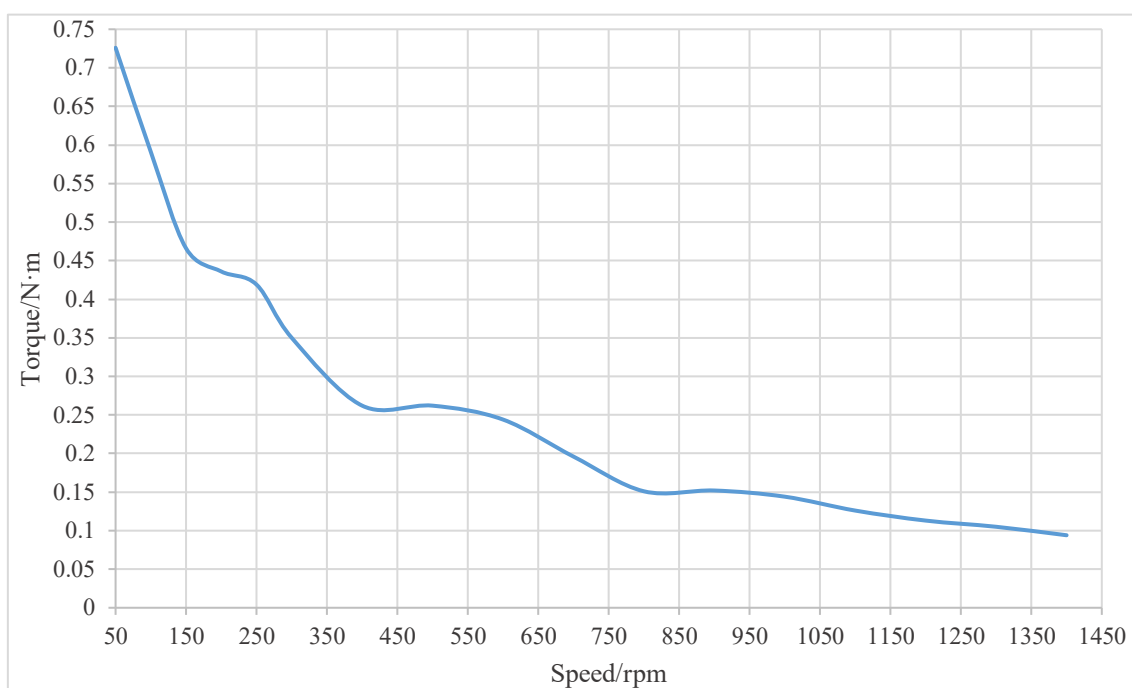


Figure7 STM4260B-CANopen/485-M-XX Frequency-Torque Characteristics(@24V, 2.0A)

3. Specifications

3.1 Technical Specifications

Table2 Motor Technical Specification

Model		STM4234B-CANopen/485-M-XX	STM4240B-CANopen/485-M-XX	STM4248B-CANopen/485-M-XX	STM4260B-CANopen/485-M-XX
The range of input power supply voltage(VDC)		12~36			
Rated voltage (VDC)		24			
Rated current (A)		1.3	1.7	1.7	2.0
Rated torque (N·m)		0.28	0.40	0.50	0.70
Working Environment	Temperature(°C)	0~40			
	Humidity (RH)	10%~85%, No condensation			
	Altitude (m)	<1000			
	Installation environment	No corrosive gas, inflammable material, oil mist, etc., and no strong vibration			
	Installation method	Horizontal or vertical			
Driver method		Using FOC magnetic field oriented control technology and SVPWM			
Encoder		Integrated 2500 lines magnetic encoder			
Communication protocol	CANopen	·CAN ports; 3-wire non-isolated; 3.3V ·CAN2.0A protocol ·Application layer: CANopen protocol ·Transmission Rate: 1Mbps,800kbps,500kbps,250kbps,125kbps, 100kbps,50kbps, 20kbps,10kbps; The default is 1Mbps.			
	RS485	·RS485 ports; 3-wire non-isolated ·Modbus RTU protocol ·Transmission Rate: 1Mbps,500kbps,256kbps,115.2kbps, 57.6kbps, 38.4kbps,19.2kbps,9.6kbps; The default is 115.2kbps.			

Control mode	CiA402	·Profile position mode (PP), Profile velocity mode(PV), Profile torque mode(PT), Velocity mode(VM) ·Interpolated position mode(IP) ·Homing mode(HM) ·Cyclic synchronous position mode(CSP), Cyclic synchronous velocity mode(CSV), Cyclic synchronous torque mode(CST)
	NiMotion	·Position mode: pulse control, multi-segment position control, digital quantity setting control ·Velocity mode: analog speed control, digital quantity setting control ·Torque mode: digital quantity setting control
Control signals	DI(Digital input)	·Quantity: 3 ·Maximum PWM pulse frequency is 2MHz ·Support for 5V and 24V level inputs ·Configurable functions: 1. Enable;2. Positive and negative limit;3. Origin switch;4. Alarm reset;5. Pause;6. Toggle multisegment run instructions.
	DO(Digital output)	·Quantity: 2 ·Output method: MOS open drain output ·Electric current: - DO1: 0.5A@36V - DO2: 1.0A@36V (Note: This DO is not available for motors with brakes.) ·Configurable functions:1. Motor running and stopping; 2. Arrival of target; 3. Alarm output.
	AI(Analog input)	·Quantity: 1 ·Input method: Single-ended input; ·Type of signal: Voltage; ·Resolution ratio: 12bit ·Voltage: 0~10V ·Precision: Quoted error≤1% ·Configurable functions: Dead band, Filter, Bias, Rate, Direction
	Fault diagnosis	·Overvoltage/Undervoltage/Overtemperature/Hardware failure /Stall/Overload/Overspeed/Initialization failure /Storage failure/Over limit detection/Homing timeout/Track failure/Target position overflow fault, etc.
	Fault reset	The alarm is self-reset, and the fault is manual reset.
	Software	The position range is set by the software, and the motor lock shaft is out of the

Features	position limit	range and the alarm is raised.
	Identification of parameters	It has the function of parameter identification and PI parameter self-tuning.
	Settable parameters	Acceleration, deceleration, jerk, speed feedforward, position feedforward, operating current, electronic gear ratio, three-ring PI parameters, following error, etc.
	Gain switching	Two groups of gain parameters can be preset, and the two groups of gain can be switched automatically according to different motor operating states.
	Resonance suppression	Effectively suppress vibration frequency range is greater than or equal to 300Hz.
	Low frequency suppression	Effectively suppress vibration frequency range is less than or equal to 100Hz.
	Parameter save and restore	To save and restore the default parameters.
	Online upgrade	Update in time according to the actual needs of the product to improve maintainability and efficiency

3.2 Indicator

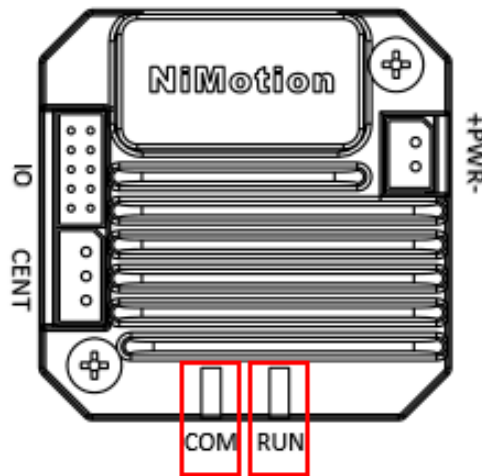


Figure8 Indicator

Table3 Meaning of Indicator

Meaning	Yellow light (COM)	Green light (RUN)
Normal communication, no alarm.	Light flashes slowly (1s flashing frequency)	Steady light
Normal communication, internal warnings are generated.	Light flashes (0.5s flashing frequency)	Steady light
Normal communication, internal faults occur.	Light flashes (0.25s flashing frequency)	Steady light
Normal communication, the hardware failure occurs.	Light off	Steady light
Abnormal communication, internal warnings are generated.	Light flashes (0.5s flashing frequency)	Light flashes (0.5s flashing frequency)
Abnormal communication, internal faults occur.	Light flashes (0.25s flashing frequency)	Light flashes (0.5s flashing frequency)
The power supply is abnormal or has a serious fault.	Light off	Light off

3.3 Definition of Terminal

3.3.1 Power Port

When the motor is placed on its side as shown in the figure below, when looking directly at the interface, the top of the power port is the first pin.

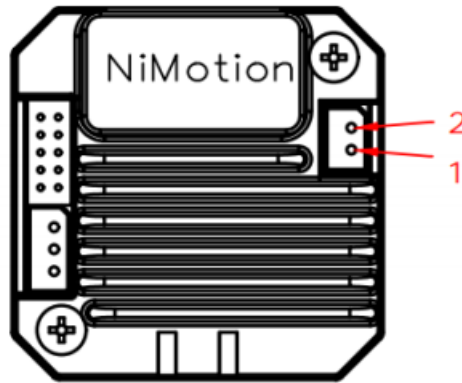


Figure9 Power Port

Table4 Definition of Power Ports

Pin	Function	Note
1	GND	-
2	VCC	12~36VDC

Note: The positive and negative poles of the power cord cannot be connected in reverse, otherwise the motor may be damaged. And the original manufacturer cannot provide warranty service for such damage.

3.3.2 Communication Port

The communication port CNET is used for the connection of RS485/CANopen communication. When the motor is placed on its side as shown in the figure below, when looking directly at the interface, the top of the CNET terminal is the first pin.

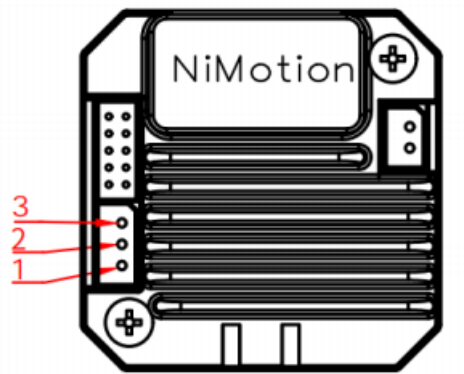


Figure10 Pins of Communication Port

Tabled5 Definition of Communication Ports

Pin	RS485	CAN
1	485 - (B)	CAN_H
2	485+ (A)	CAN_L
3	GND	GND

3.3.3 Input and Output

input and output connection ports.

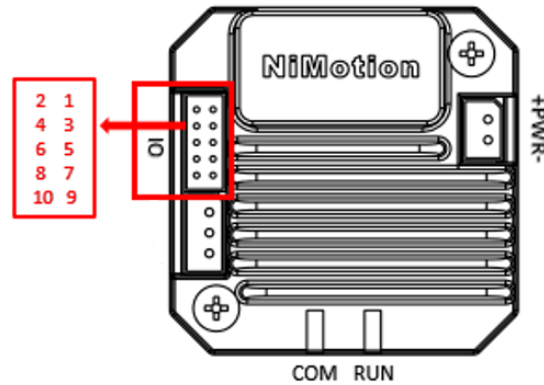


Figure11 Pins of I/O Port

Table6 Definition of I/O Ports(Viewed from butt joint)

Pin	Meaning	Function
1	DI1	Support for 5V and 24V level input All adopt high-speed bidirectional optical coupler, support PWM
2	+5V_COM	+ 5V input common terminal
3	DI2	Support for 5V and 24V level input All adopt high-speed bidirectional optical coupler, support PWM
4	+24V_COM	+ 24V input common terminal
5	DI3	Support for 5V and 24V level input All adopt high-speed bidirectional optical coupler, support PWM
6	AI	Analog input ranges from 0 to10V
7	DO1	Output method: Support MOS open-drain output Current: 0.5A@36V
8	GND	Ground
9	DO2	Output method: Support MOS open-drain output Current: 1.0A@36V Note: For the motor with brake , the DO is not available.
10	-	Reserved

Note: The input signal type of DI direct level signals, such as logical level signals output by devices such as signal generators, are not supported, and only PNP, NPN, and dry contact signals are supported.

3.3.4 Wiring Diagram of Communication

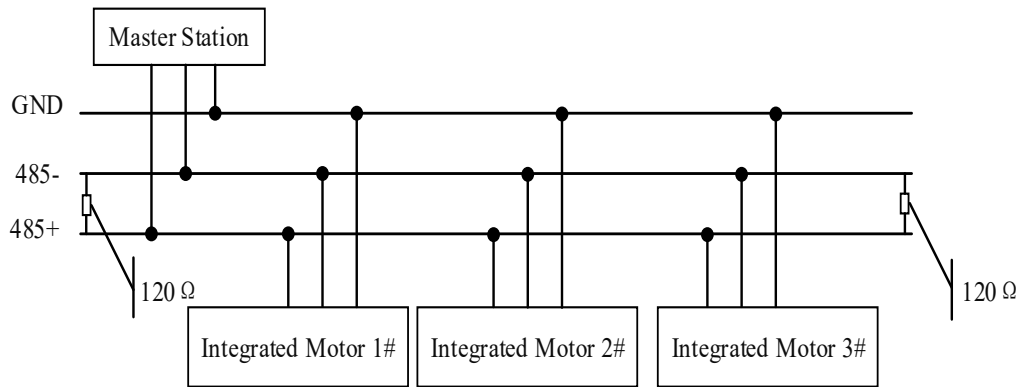


Figure12 Wiring Diagram of RS485 Communication

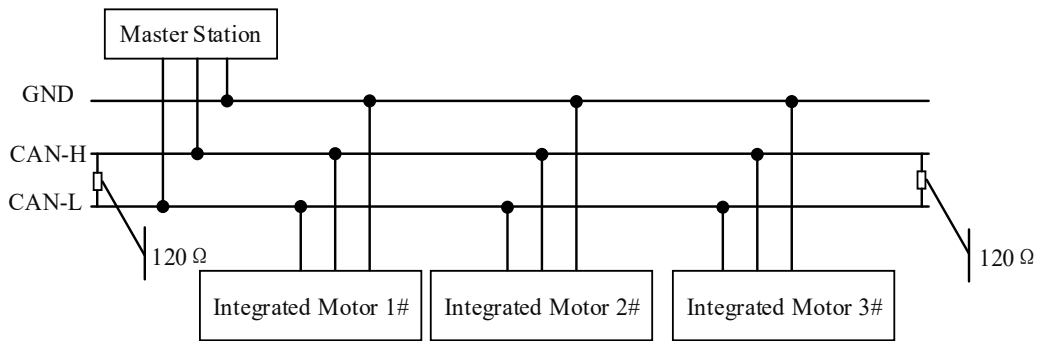


Figure13 Wiring Diagram of CANopen Communication

3.3.5 Wiring Diagram of I/O

DI1, DI2, DI3 (input terminals) are connected to the NPN sensor.

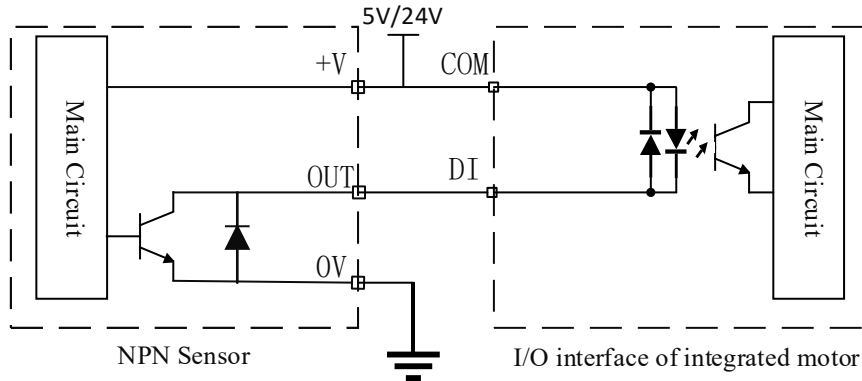


Figure14 Schematic diagram of DI input port is connected to NPN sensor

+V terminal of the sensor and the COM terminal of the I / O interface are connected to the positive 5V / 24V power supply, 0V terminal of the sensor is connected to the negative power supply, and DI terminal of the I / O interface is connected to OUT terminal of the sensor.

DI1, DI2, DI3 (input terminals) are connected to the PNP sensor.

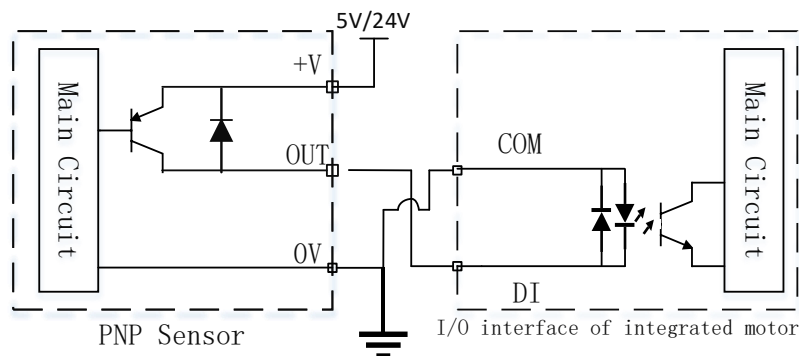


Figure15 Schematic diagram of DI input port is connected to PNP sensor

The 5V / 24V positive pole of the power supply is connected to +V terminal of the sensor, 0V terminal of the sensor is connected to the negative pole of the COM terminal of the I / O interface, and OUT terminal of the sensor is connected to DI terminal of the I / O interface.

DI1, DI2, DI3 (input terminals) are connected to the switch (Common-anode).

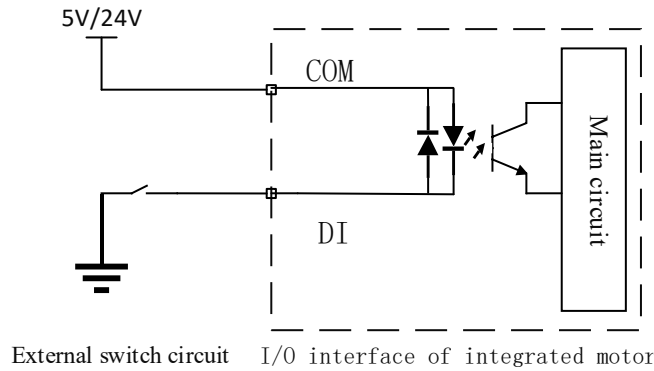


Figure16 Schematic diagram of DI input port is connected to switch

The 5V / 24V positive pole of the power supply is connected to COM terminal of the I / O interface, one end of the switch is connected to DI terminal of the I / O interface, and the other end of the switch is connected to the negative pole of the power supply.

DI1, DI2, DI3 (input terminals) are connected to the switch(Common cathode).

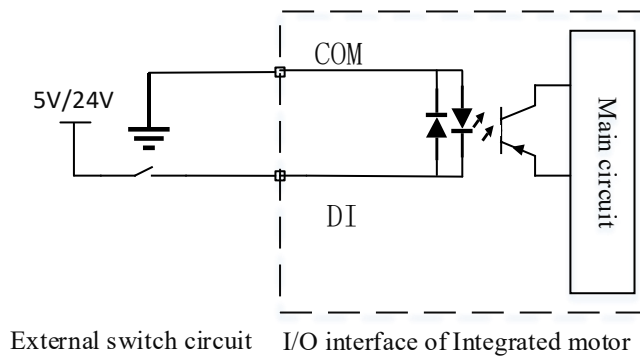


Figure17 Schematic diagram of DI input port is connected to switch

The 5V / 24V positive pole of the power supply is connected to one end of the switch, the other end of the switch is connected to DI terminal of the I / O interface, and COM terminal of the I / O interface is connected to the negative pole of the power supply.

DI1, DI2, DI3 (input terminals) are connected to pulse.

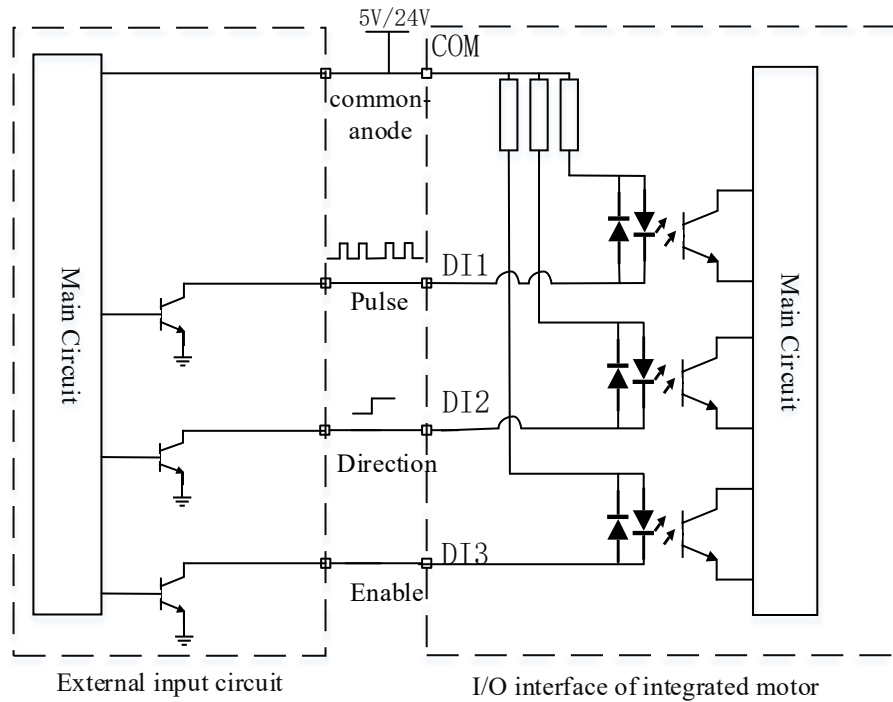


Figure18 Schematic diagram of DI input port is connected to pulse

DO1 (output terminal) is connected to the LED.

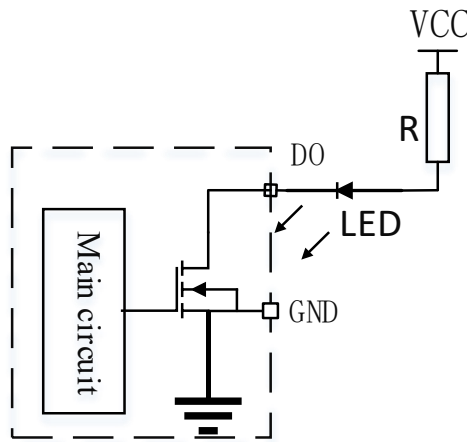


Figure19 Schematic diagram of DO Port is connected to the LED

The positive terminal of the power supply VCC is connected to the positive electrode of the LED lamp after the current limiting resistor, DO terminal of the I / O interface is connected to the negative electrode of the LED lamp, and GND terminal of the I / O interface is connected to the negative electrode of the power supply.

AI port is connected to analog input.

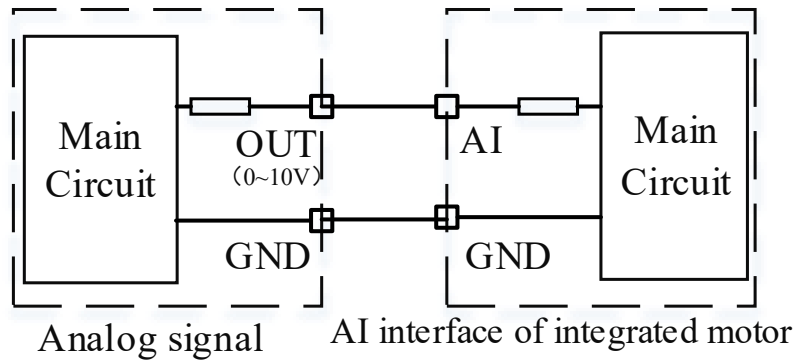


Figure20 Schematic diagram of AI port is connected to analog signal

AI terminal of integrated motor is connected to the OUT terminal of the analog signal, and the GND of the motor is connected to the GND of the analog signal.

4. Alarm Function

Table7 Table of Alarm Function

Alarm Code	Alarm Contents	Default Alarm Type	Self reset or not
0x2300	Motor overcurrent alarm	Fault	No
0x4012311	Motor overload alarm	Alarm	Yes
0x3002312	Motor stalled alarm	Pause and lock the shaft	No
0x1013210	Power supply overvoltage alarm	Fault	Yes
0x1013220	Power undervoltage alarm	Fault	Yes
0x14210	Overtemperature alarm	Fault	Yes
0x14220	Low temperature alarm	Fault	Yes
0x5080	Drive failure	Fault	No
0x5540	Flash operation failure	Fault	No
0x5541	Flash initialization failure	Fault	No
0x4005542	Flash validation error warning	Alarm	No
0x5543	No parameters in flash user area	Fault	No
0x6000	Hardware initialization failure	Fault	No
0x16320	Parameter setting error	Fault	Yes
0x6321	Registration failure	Fault	No
0x7305	Z pulse fault	Fault	No
0x7308	Encoder fault	Fault	No
0x4017307	Encoder warning	Alarm	Yes
0x4017308	Encoder calibration failure	Alarm	Yes
0x17310	Overspeed	Fault	Yes
0x3017501	Communication failure	Alarm	Yes
0x4017502	Illegal address	Alarm	Yes
0x4017503	Illegal data value	Alarm	Yes
0x4017505	Confirmation in communication	Alarm	Yes
0x4017506	Slave in communication	Alarm	Yes
0x401750C	Request data is larger than mapping data	Alarm	Yes
0x401750D	The number of request data is not equal to the number of mapping data	Alarm	Yes
0x401750E	The address of the packet node is incorrect	Alarm	Yes
0x8610	The origin regression timeout	Fault	No
0x8611	Out of position	Fault	No
0x3018613	Software limit error	Pause and lock the shaft	No

Alarm Code	Alarm Contents	Default Alarm Type	Self reset or not
0x3018614	The limit switch is incorrect	Pause and lock the shaft	No
0x4018615	Curve planning calculation error	Alarm	Yes
0x18616	Target location overflow	Fault	No
0x5018617	Curve planning parameters are too small	ignore	Yes
0xFF01	Motor parameter identification failure	Fault	No
0x0401FF02	Parameter saving fault	Alarm	Yes
Note: Please refer to the corresponding communication manual for more details of alarm failure.			

5. Daily Inspection and Maintenance

Table8 Daily Inspection and Maintenance

Inspection Items	Inspection Time	Key points of inspection and maintenance	Note
Vibration and sound inspection	Everyday	Visual and auditory judgment	-
Fixing and connecting	At least once a week	Periodically check whether the screws at the motor installation site, the motor rotor and the mechanical connection, the screws at the terminal station and the mechanical part are loose.	-
Inspection of appearance	Once a week	Wipe with cloth or clean with air gun.	-
Comprehensive inspection	At least once every 20,000 hours or every 5 years	-	-

- It is forbidden to reprint or copy all or part of this manual.
- Product performance, specifications and appearance are subject to change without prior notice due to improvement, please forgive me.
- We strive to make the contents of the manual as accurate as possible. If you find any problems, errors or omissions, please contact Beijing NiMotion Control Technology Co., Ltd.

Beijing NiMotion Control Technology Co., Ltd.
Building 3, No. 12 Jinxing Road, Daxing District, Beijing
Postcode: 102628
Tel: (010)60213882 Fax: (010)60213882
Mail: nimotion@nimotion.com
Website: <http://www.nimotion.com>